

S.3 HOLIDAY WORK ACIDS AND AKALIS

The human stomach contains hydrochloric acid with a pH of about 2. The acid helps to kill any germs in our stomach. However, when the stomach walls produce too much of the acid, we suffer from stomach pains and heartburn.



One way to deal with this problem is to take an antacid. The main component of antacid is magnesium tablets that contain magnesium hydroxide
Explain how the antacid helps to cure heartburn.

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Write a word equation for the reaction between magnesium hydroxide and hydrochloric acid

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State the kind of reaction that occurred above.

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When food particles remain in the teeth after a meal, the sugars in them attract bacteria which through fermentation processes may lead to tooth decay.



One way of preventing this is by regularly brushing teeth using toothpaste. The toothpaste is always alkaline in nature. Give an explanation for this.

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ATTEMPT ONE OF THE PROJECTS

PROJECT 1: PREPARATION OF INDICATORS FROM RED CABBAGE



By the end of this activity, you should be able to:

- Prepare an indicator from red cabbage.
- Test its colour in acid and base.

What you need

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|-----------------------------|-----------------------------|--------------------|
| • 250 ml beakers | • hot plates | • beaker tongs |
| • 2l distilled water | • large screw | top storage bottle |
| • dropper bottles of 1M HCl | • dropper bottle of 1M NaOH | |
| • red cabbage | • watch glasses goggles | |

What to do

1. Pull a few leaves of red cabbage into very small chunks with your fingers.
2. Fill a 250 ml beaker halfway with torn cabbage leaves and add water until the beaker is about two-thirds full. Place the beaker on a hot-plate at a medium temperature.
3. You should notice that the water turns purple as the cabbage boils. When the water is very purple, remove the beaker from the heat with beaker tongs and let cool. The purple juice you made is acid-base indicator.
4. To find the colour of your indicator in acid, add one drop of indicator to one drop of 1M Hydrochloric acid in a watch glass. Record the colour of the indicator in acid.
5. To find the colour of your indicator in acid, add one drop of indicator to one drop of 1M sodium hydroxide in a watch glass. Record the colour of the indicator in a base.

PROJECT 2: PREPARATION OF INDICATORS FROM PLANT

MATERIALS



By the end of this activity, you should be able to

- Prepare an indicator from flower petals

What you need

- glass beakers (50cm³)
- filter papers
- glass rod
- test tubes
- red or blue flower petals
- ethanol or local waragi
- dilute sulphuric acid
- dilute sodium hydroxide
- filter funnel
- stand and clamp
- mortar and pestle
- measuring cylinder (5 cm³)
- water
- lemon or orange juices
- soap solution
- solution of ashes (from plants) small bottles

What to do

1. Collect blue flowers from the compound and then follow the steps as given in 2 – 8 below.
2. Place the blue flower petals in a mortar.
3. Grind the petals using a pestle.
4. Add a portion of a solvent made by mixing ethanol (Waragi) with water and continue grinding until the extract is as deep in colour as possible.
5. Decant the extract into a test tube until you have about 10cm³ of it.
6. Repeat procedure 1 – 4 with red flower petals.
7. Use the flower extracts to test the solutions: lemon or orange juices dilute sulphuric acid, soap solution, dilute sodium hydroxide, solution of ashes (from plants) and soap solution.
8. Record your observations in a suitable table and discuss them with your fellow students. Classify the solutions as acids and bases.

OBSERVATIONS

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Activity of Integration(SALTS)

JUDITH a senior 2 student was in the lab reading books. The lab technician came in and asked her which substance can be used in the lab to detect for presence of water. She proudly replied and said it was anhydrous copper (II) sulphate powder which is a white solid. She continued and said that it turns to a blue crystal when it absorbs water. she requested the lab technician to help her carry out an experiment to detect whether air contains water vapor but unfortunately the anhydrous copper (II) sulphate powder and copper (II) sulphate crystals were not available in the lab. The lab technician tasked Judith to use the existing copper (II) oxide powder which is a black solid that was around and a suitable dilute acid to make her own copper (II) sulphate crystals, which she can later dehydrate gently with heat to obtain her anhydrous copper (II) sulphate powder. Unfortunately, Judith couldn't proceed with the experiment.



TASK. As a senior 2 student who has studied the concepts of salt preparation, help Judith prepare copper (II) sulphate crystals that she can dehydrate and test for presence of water vapor in the atmosphere.